

Task 12 — Pattern Analysis

Equivalence-class discovery in Form Space

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Track	Algorithm-only, unclassified, laptop-based evaluation
Solicitation	W91RUSI-FCID260001 (SUB-007)
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Handling	UNCLASSIFIED // PROPRIETARY
Version	FormSpace overlay v1.0 — 2026-09-01

1. Overview — Pattern as Class Discovery in Form Space

Pattern Analysis is the operation of identifying Form-equivalence classes over an ordered stream of multi-modal observations. Given the learned signal map $\Lambda: X \rightarrow Z$ and the quotient $F_\Lambda = X/\sim_\Lambda$, a pattern is a temporally coherent cluster of observations that map to a common region of F_Λ . The Nature Learning Model estimates the posterior over latent state $q_\theta(s_t | o_{1:t})$; the pattern score is the log-likelihood that consecutive windows share an equivalence class under the learned tolerance.

2. Method — Selective State-Space Backbone

We use a selective state-space (SSM/Mamba-style) backbone with input-dependent Δt , B , C parameters. The backbone runs at three tier sizes: 120 M (edge), 500 M (ITDX-preferred), and 1.5 B (stretch). Training objectives include masked reconstruction, contrastive alignment across modalities, physics regularization, calibration (Brier + Expected Calibration Error), graph consistency, causal invariance, and a composite safety loss (10 loss terms total; see Evaluation Harness).

Two baselines run before the learned model in every evaluation:

- Deterministic: robust z-score with median/MAD normalization plus persistence window (see Coastal Sentinel §8.1 for the equation).
- Modern sequence baseline: matched-data-split transformer at equivalent parameter count.

3. Inputs

- Rooted observations per canonical schema (identity, time, uncertainty, location, uncertainty, modality, calibrated features, quality, raw ref, hash).
- Device-health and link-state streams from the DIRTNet 4-node emulator.
- Controlled-stimulus truth markers with disclosed labels.
- Optional human adjudication where available.

4. Outputs

- Anomaly score per observation and per window.
- Ranked Form-class candidates with confidence.
- Time-to-detection against truth markers.

- Calibrated confidence with uncertainty decomposition (aleatoric + epistemic).
- Evidence IDs and Merkle-verified lineage.
- Explicit missing-modality flags and calibration age.

5. Field Ablations (run on every evaluation)

- Remove FCI (bioelectric) channel.
- Remove Agaric imagery.
- Remove Psathyrella acoustic context.
- Inject 250 ms clock error.
- Inject 5% packet delay.
- Inject linear sensor drift.
- Compare centralized-raw versus edge-features-plus-signed-provenance.

6. Metrics

- Precision / recall / F1 with grouped bootstrap over 5 seeds {11,17,23,29,31}.
- Time-to-detection distribution (median + 95% CI).
- Expected Calibration Error (ECE) and Brier score.
- Class-stability across seeds — do the same equivalence classes recover under seed variation?
- Node-dropout degradation curve.
- Conformal prediction interval width at $\alpha = 0.1$.

7. Timing Budget (live, on-site)

- Cold start ≤ 30 s.
- Live inference ≤ 90 s total on the on-site sample.
- Memory gate: 80% VRAM ceiling triggers automatic tier downgrade with visible UI label.

8. Failure Modes and Fallback Ladder

The fallback ladder is: medium $\rightarrow$ small $\rightarrow$ quantized $\rightarrow$ distilled $\rightarrow$ classical $\rightarrow$ prerecorded $\rightarrow$ static. Every fallback state is labeled in the operator UI so an Army SME can tell exactly which mode is running. A single drifting sensor is expected to shift the Form-class boundary; NLM must downgrade confidence and flag calibration rather than silently accept the drifted stream.

9. Underperformance Statement

If the live configuration does not meet its preregistered target, the PI states on the record: the observed metric, the preregistered target, the deterministic-baseline value, and that the run is signed and reproducible. We do not attempt to fix failing runs live.

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